The chain-strength prefactor, the coupler range, and shimming

Why p must grow with N, where it stops, and how the machine's biases are measured and corrected

`Projects_master_HPQC/scripts/QuantumComputing/Dimod`

Abstract

The QPU solves the SK planner problem only after embedding it: each agent becomes a chain of qubits held together by a ferromagnetic coupling $c = p\,J_{\mathrm{rms}}\sqrt{N-1}$. This note derives, step by step, a *lower* bound on p (the chains must survive the field of the other $N-1$ agents) and an *upper* bound (after the QPU rescales the problem into its coupler range, the SK couplings must stay above the analogue control error). The lower bound measured on the campaign's instances is $p \approx 3$–4 at the N that matter, *above* the previous default 1.414: the intuition that a larger value ("$J = 3$") is needed is correct. The upper bound decreases as $1/\sqrt{N-1}$, so the window closes near $N \approx 100$ unless the extended coupler range is used. The second half describes the flux-bias shimming of D-Wave's tutorial, adapted to SK, as implemented in `shimming.py`, and what it measures about the state of the machine.

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1 Setting and notation

The planner's problem for N agents is the SK Hamiltonian without field,

$$H(S) = -\sum_{i<j} W_{ij} S_i S_j, \qquad S_i = \pm 1, \qquad W = \tfrac{1}{2}(J + J^{\mathsf{T}}),\ W_{ii} = 0. \tag{1}$$

For the pure spin glass ($\varepsilon = 0$, $J_0 = 0$) the entries are independent Gaussians,

$$W_{ij} \sim \mathcal{N}\big(0, J_{\mathrm{rms}}^2\big)\,, \qquad J_{\mathrm{rms}} = \frac{1}{\sqrt{N}} \qquad (\texttt{InteractionEnsemble},\ \sigma = 1). \tag{2}$$

Local field. The field felt by agent i in configuration S is

$$h_i(S) = \sum_{j \neq i} W_{ij} S_j, \qquad H(S) = -\tfrac{1}{2} \sum_i S_i h_i(S). \tag{3}$$

For a fixed S independent of W it is a sum of $N-1$ independent terms of variance J_{rms}^2:

$$\mathbb{E}\, h_i^2 = (N-1) J_{\mathrm{rms}}^2 \quad \Longrightarrow \quad |h_i| \sim J_{\mathrm{rms}} \sqrt{N-1} = \sqrt{\tfrac{N-1}{N}} \approx 1. \tag{4}$$

This is the natural unit for the chain coupling, hence the definition used by the code (`uniform_torque_compensation`):

$$\boxed{c = p\, J_{\mathrm{rms}} \sqrt{N-1}} \qquad \Longrightarrow \qquad p = \frac{c}{\text{typical } |h_i|}. \tag{5}$$

For our ensemble $c = p\sqrt{(N-1)/N} \approx p$.

Embedding. Agent i is a chain C_i of L_i qubits (a tree on the chip graph, $L_i \approx N/12 + 1$ on Zephyr, $N/12 + 2$ on Pegasus for the clique embeddings). Inside the chain each edge carries the Ising coupling $-c$. Between chains i and j there are $|E_{ij}| \geq 1$ physical couplers (1.1–1.2 on average), and W_{ij} is split evenly over them. The physical problem is

$$H_{\mathrm{phys}}(s) = -\sum_{i<j} \sum_{(q,q') \in E_{ij}} \frac{W_{ij}}{|E_{ij}|} s_q s_{q'} \;-\; c \sum_i \sum_{(q,q') \in C_i} s_q s_{q'}. \tag{6}$$

2 Lower bound: the chains must hold

2.1 When does a chain break?

Take chain i, all other chains intact. The rest of the problem acts on chain i as a field h_i, distributed over its qubits: qubit $q \in C_i$ receives $h_{i,q}$, with $\sum_{q \in C_i} h_{i,q} = h_i$. Suppose the chain points against its field, $S_i h_i < 0$ (it does so for a while during the anneal, and in every excited state). Flipping a connected segment $A \subset C_i$ that is cut from the rest by $b(A)$ chain edges changes the energy by

$$\Delta E(A) = \underbrace{2c\, b(A)}_{\text{broken chain edges}} \;-\; \underbrace{2\Big|\sum_{q \in A} h_{i,q}\Big|}_{\text{field energy gained}}. \tag{7}$$

The segment that breaks most easily is an end segment of the tree, $b(A) = 1$, carrying almost all of the field. The chain is stable against every segment flip if

$$c \;\geq\; \max_A \Big|\sum_{q \in A} h_{i,q}\Big| \;\lesssim\; |h_i|, \tag{8}$$

so a sufficient condition for chain i is $c \geq |h_i|$, and for the whole embedding

$$\boxed{c \;\geq\; \max_i |h_i(S)| \quad \text{for the configurations } S \text{ the anneal visits near its end.}} \tag{9}$$

At a finite effective temperature T_{eff} at freeze-out, a chain with margin $c - |h_i|$ still breaks with probability $\sim e^{-2(c-|h_i|)/T_{\mathrm{eff}}}$: (9) is the scale, not a sharp threshold. In units of $J_{\mathrm{rms}}\sqrt{N-1}$ the condition reads $p \geq p_*(S) = \max_i |h_i(S)|/(J_{\mathrm{rms}}\sqrt{N-1})$.

2.2 Three estimates of the field

(a) Typical field, p_{typ}. For random S, $|h_i|/(J_{\text{rms}}\sqrt{N-1})$ is a standard half-normal, with rms 1. At a low-energy state the fields are aligned with the spins and larger: measured at greedy local minima, the rms is $p_{\text{typ}} = 1.39$ at $N = 16$, 1.50 at $N = 64$, 1.50 at $N = 104$.

(b) Largest field at a low-energy state, p_{loc}. What (9) needs is the *maximum over the N chains*. At a local minimum $p_{\text{loc}} = \max_i |h_i|/(J_{\text{rms}}\sqrt{N-1})$ grows slowly with N (extreme value of N roughly Gaussian variables, $\sim \sqrt{2\ln N}$):

N	16	64	104	160
p_{typ} (rms field)	1.39	1.50	1.50	1.52
p_{loc} (largest field, local minimum)	2.66	3.28	3.41	3.70
p_{wc} (worst case, $\max_i \sum_j \lvert W_{ij}\rvert$)	4.22	7.74	9.63	11.72
$c/\max\lvert W_{ij}\rvert$ at $p = 1.414$	1.93	3.08	3.72	4.42

(40 instances per N, script `prefactor_figure.py`.)

(c) Worst case, p_{wc}. Over all configurations, $|h_i| \le \sum_j |W_{ij}|$, reached when every neighbour pulls the same way. With $\mathbb{E}|W_{ij}| = \sqrt{2/\pi}\, J_{\text{rms}}$,

$$\sum_{j\neq i} |W_{ij}| \approx \sqrt{\tfrac{2}{\pi}}\,(N-1)J_{\text{rms}} \quad\Longrightarrow\quad p_{\text{wc}} \approx \sqrt{\tfrac{2}{\pi}}\sqrt{N-1}\ \text{(mean over } i\text{)};\ \text{the max over } i \text{ is a bit larger.} \tag{10}$$

Numerically $\sqrt{2/\pi}\sqrt{N-1} = 0.798 \times \sqrt{103} = 0.798 \times 10.15 = 8.10$ at $N = 104$. This bound guarantees that no configuration breaks a chain. It is far too pessimistic: such configurations have high energy and are not where the anneal ends.

Conclusion of the lower bound. The relevant scale is p_{loc}:

$$p \gtrsim p_{\text{loc}} \approx 3 \ \ (N \approx 30\text{–}60), \qquad \approx 3.5 \ \ (N \approx 100\text{–}180). \tag{11}$$

The previous default $p = 1.414$ is below this for every $N \ge 16$: in low-energy states the most stressed chain is held by less than half of what it needs. Majority vote then repairs the broken chain, but the read is biased. A grid that tests p up to 6 is needed, and `coupling-scan` now uses $0.75, 1, 1.414, 2, 3, 4, 6$.

2.3 What "$J = 3$" means here

Two readings are common, and both point the same way.

1. *Chain coupling three times the largest problem coupling*, $c = 3\max|W_{ij}|$. The largest of the $N(N-1)/2$ Gaussians is $\max|W_{ij}| \approx J_{\text{rms}}\sqrt{2\ln(N(N-1))}$; at $N = 104$: $\sqrt{2\ln 10712} = \sqrt{2 \times 9.28} = 4.31$, so $\max|W| \approx 4.31/\sqrt{104} = 0.42$ and $c/\max|W| = 1.414\times\sqrt{103/104}/0.42 \approx 3.3$ (measured: 3.72). In this reading the old default was already at "3" for $N \gtrsim 64$, but below it for small N (1.93 at $N = 16$).

2. *Chain coupling ≈ 3 in units of the typical field*, i.e. $p = 3$. This is the p_{loc} scale derived above.

The second reading is the one tied to the physics of chain breaking (fields, not single couplings); for comparison, Teza et al. quote chain couplings of 15–20 times the typical problem coupling; here $c/J_{\text{rms}} = p\sqrt{N-1}$ is $1.414 \times 10.15 = 14$ at $N = 104$ for the old default and $3 \times 10.15 = 30$ for $p = 3$. Their problem is not all-to-all, so the fields on their chains are smaller: the field argument, not the ratio, is what transfers.

3 Upper bound: the SK couplings must stay above the noise

3.1 Compression by the coupler range

The QPU accepts couplings in $j_{\text{range}} = [-1, 1]$. With `auto_scale` it divides the whole physical problem by

$$D = \max\left(c,\ \max_{i<j} \frac{|W_{ij}|}{|E_{ij}|}\right) = c \qquad \text{as soon as } c \geq \max|W_{ij}| \ (p \gtrsim 0.45 \text{ at } N = 104,\ p \gtrsim 0.75 \text{ at } N = 16). \tag{12}$$

The SK coupling W_{ij} is then programmed as W_{ij}/D in total, so its rms on the chip is

$$\frac{J_{\text{rms}}}{D} = \frac{J_{\text{rms}}}{p\, J_{\text{rms}}\sqrt{N-1}} = \boxed{\frac{1}{p\sqrt{N-1}}}. \tag{13}$$

Numerically at $N = 104$: $1/(1.414 \times 10.15) = 0.070$ for $p = 1.414$; $1/(3 \times 10.15) = 0.033$ for $p = 3$; $1/(6 \times 10.15) = 0.016$ for $p = 6$.

3.2 Control error

Every programmed coupling is realised with an error ("integrated control error", ICE): $J_{qq'} \to J_{qq'} + \delta J_{qq'}$, with δJ of order $\sigma \approx 0.01$–0.03 of the range, partly static (calibration, crosstalk), partly fluctuating between programmings. The logical coupling receives the sum of $|E_{ij}|$ such errors, of size $\sigma\sqrt{|E_{ij}|}$. The relative error on each SK coupling is

$$\varepsilon = \frac{\sigma\sqrt{|E_{ij}|}}{J_{\text{rms}}/D} = p\,\sigma\sqrt{|E_{ij}|}\sqrt{N-1}. \tag{14}$$

Requiring $\varepsilon \leq 1$ (the instance on the chip still resembles the one we want) gives

$$\boxed{p \ \leq\ p_{\text{noise}} = \frac{1}{\sigma\sqrt{|E|}\sqrt{N-1}}} \approx \frac{1}{\sigma\sqrt{N-1}}. \tag{15}$$

$\varepsilon \leq 1$ is necessary, not sufficient: the SK ground state is fragile under small random changes of the couplings ("bond chaos"), so finding the *exact* ground state of the intended instance needs $\varepsilon \ll 1$. Numbers at $N = 104$ ($\sqrt{N-1} = 10.15$):

σ	0.01	0.02	0.03
p_{noise}, range $[-1, 1]$	$1/(0.01 \times 10.15) = 9.9$	4.9	3.3
p_{noise}, extended range (Sec. 5)	19.7	9.9	6.6

3.3 Field errors grow with p too

A constant flux offset on qubit q acts as a spurious field δh_q (of order $\sigma_h \sim 0.01$–0.05 of h_{range} on the programmed scale). On a chain of L qubits the logical spin receives $\sum_{q \in C_i} \delta h_q$, between $\sqrt{L}\,\sigma_h$ (random signs) and $L\,\sigma_h$ (same sign). Compared with the programmed field scale $J_{\text{rms}}\sqrt{N-1}/D = 1/p$, the relative bias is

$$\beta_i \simeq p\,\sqrt{L_i}\,\sigma_h \quad \text{to} \quad p\,L_i\,\sigma_h. \tag{16}$$

It grows with p and with the chain length, and it breaks the $S \to -S$ symmetry of the problem. This is exactly what flux-bias shimming removes (Sec. 6).

4 The window, and what it implies

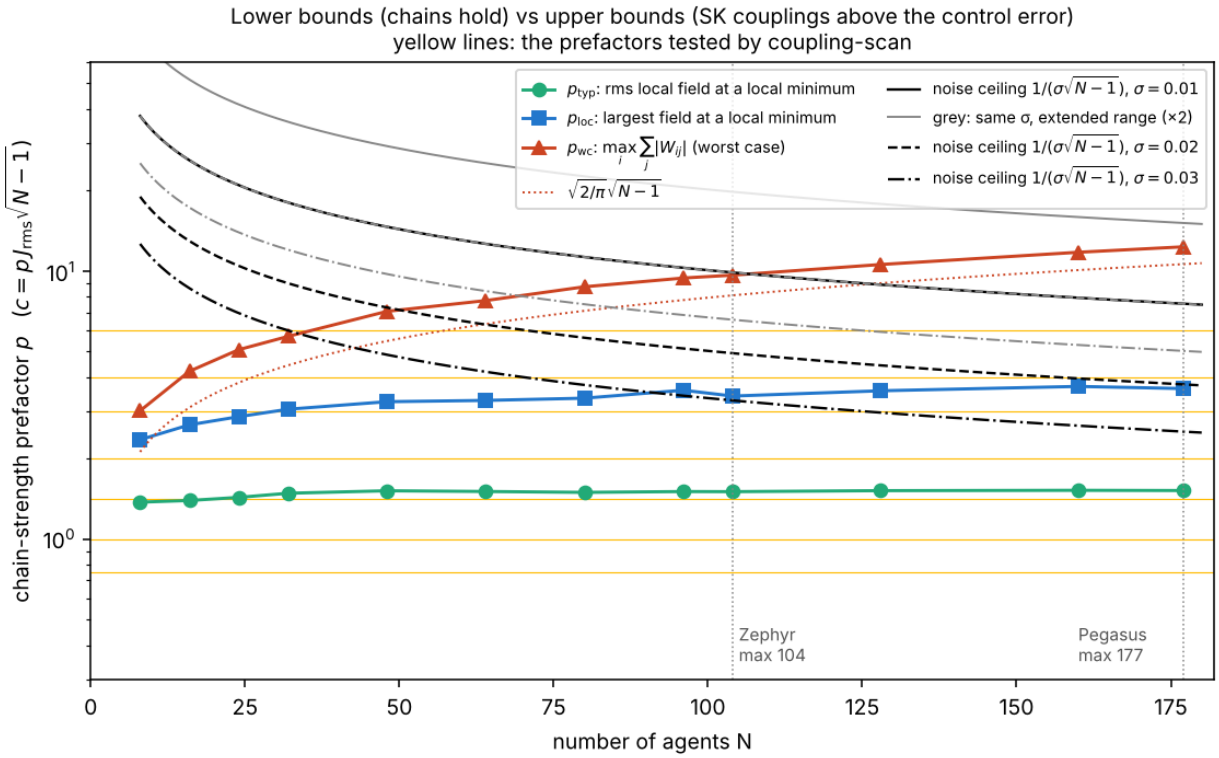


Figure 1: Lower bounds (coloured: the chains hold) and upper bounds (black: the SK couplings are above the control error σ; grey: the same with the extended coupler range). Yellow: the prefactors tested by `coupling-scan`.

The usable window is $p_{\text{loc}} \lesssim p \lesssim p_{\text{noise}}$. Setting $p_{\text{loc}} \approx 3.5$ equal to $1/(\sigma\sqrt{N-1})$ gives the size where it closes:

$$\sqrt{N_* - 1} = \frac{1}{3.5\,\sigma} \implies N_* = 1 + \frac{1}{12.25\,\sigma^2} : \quad \sigma = 0.01 \to N_* \approx 820;\ \ \sigma = 0.02 \to 205;\ \ \sigma = 0.03 \to 92. \tag{17}$$

(With the worst-case bound instead, $\sqrt{2/\pi}\sqrt{N-1} = 1/(\sigma\sqrt{N-1})$ gives $N_* = 1 + 1/(0.798\,\sigma)$: ≈ 126 for $\sigma = 0.01$, 64 for $\sigma = 0.02$.) So:

- for $N \lesssim 50$ any p in $[3, 6]$ is safe on both sides;
- at $N = 104$ on Zephyr the window is $[3.4,\ 3.3\text{–}10]$ depending on the real σ: it can be narrow, and only the scan tells;
- at $N = 128$–177 on Pegasus the extended range (if offered) doubles the room;
- the "smallest p with no broken chain" rule of `coupling-scan` picks the lower edge of the window, which is where the SK problem is least compressed.

5 Extended coupler range

Advantage QPUs report `extended_j_range` $= [-2, 1]$: ferromagnetic couplings down to -2, with a constraint on the sum of the couplings at each qubit (`per_qubit_coupling_range`). Chain couplings are ferromagnetic, the SK couplings stay in $[-1, 1]$, so the divisor becomes

$$D_{\text{ext}} = \max\Big(\frac{c}{2},\ \max_{W/|E|>0} \frac{W_{ij}/|E_{ij}|}{1},\ \max_{W/|E|<0} \frac{W_{ij}/|E_{ij}|}{-1}\Big), \tag{18}$$

and the SK couplings are programmed twice as large: p_{noise} doubles. The code (`backends.py`, `_scale`) computes D_{ext} itself, sends the scaled problem with `auto_scale=False`, checks the per-qubit sum, and falls back to `auto_scale` when the solver has no extended range or the check fails. Every result records the mode and the divisor (`info.programmed.scaling`); `validate` recomputes it.

```
python main_simulation.py init --preset full --out results/full3 --extended-j-range
```

D-Wave notes that couplings in the extended part can be less precise; the scan with and without the option at the same N is the way to see whether it helps.

6 Shimming: measuring and correcting the machine's biases

6.1 The idea (D-Wave's shimming tutorial)

The tutorial (https://github.com/dwavesystems/shimming-tutorial, Chern et al., Front. Comput. Sci. 2023) refines the calibration for a specific problem by using a *symmetry* of that problem: quantities that the symmetry forces to be equal are measured, and the deviations are fed back into small corrections. Two kinds of correction exist:

- **flux-bias offsets** Φ_q per qubit (parameter `flux_biases`, units of Φ_0), which cancel spurious fields; the target is a magnetisation that the symmetry fixes;
- **coupler adjustments**, which equalise the frustration of couplers that the symmetry makes equivalent ("orbits").

6.2 Why flux shimming applies to SK, and coupler shimming does not

With $h = 0$, $H(S) = H(-S)$. Every Boltzmann or quantum state of the ideal Hamiltonian is symmetric, so for every physical qubit

$$\langle s_q \rangle = 0 \qquad \text{exactly.} \tag{19}$$

Any measured $m_q = \frac{1}{R}\sum_{r=1}^{R} s_q^{(r)} \neq 0$ beyond statistical error is a device bias. Random SK couplings have no symmetry relating different couplers, so there are no orbits and coupler shimming has no target; it is not used.

6.3 The feedback rule and its convergence

For small offsets the response is linear: $m_q \approx \chi_q\,(\Phi_q^{\text{dev}} + \Phi_q)$, with Φ_q^{dev} the unknown device offset and $\chi_q > 0$ a susceptibility. The update

$$\Phi_q^{(t+1)} = \Phi_q^{(t)} - \alpha_\Phi\, m_q^{(t)} \qquad \text{(tutorial; default } \alpha_\Phi = 10^{-5}\,\Phi_0) \tag{20}$$

gives for the residual $e^{(t)} = \Phi^{\text{dev}} + \Phi^{(t)}$

$$e^{(t+1)} = (1 - \alpha_\Phi \chi)\, e^{(t)} - \alpha_\Phi\, \eta^{(t)}, \tag{21}$$

where η is the shot noise of m_q. It converges geometrically if $0 < \alpha_\Phi\chi < 2$, with time constant $1/(\alpha_\Phi\chi)$ iterations. At stationarity the residual jitters with variance

$$\operatorname{Var} e = \frac{\alpha_\Phi^2 \operatorname{Var}\eta}{1 - (1 - \alpha_\Phi\chi)^2} \approx \frac{\alpha_\Phi \operatorname{Var}\eta}{2\chi} \qquad (\alpha_\Phi\chi \ll 1)\,: \tag{22}$$

a small α_Φ averages more reads, a large one converges faster.

The shot-noise floor. In a low-energy SK state each read picks one of the two mirror states, so $s_q = \pm 1$ with equal probability and

$$\operatorname{Var} m_q = \frac{1}{R} \implies \text{rms } m_q \approx \frac{1}{\sqrt{R}} = 0.1 \text{ for } R = 100. \tag{23}$$

A measured rms $|m_q|$ near $1/\sqrt{R}$ is *noise, not bias*; only the excess above it is device bias. Over T iterations the offsets average TR reads, so biases down to $\sim 1/\sqrt{TR}$ ($= 0.018$ for $T = 30$, $R = 100$) are corrected. `shim_history.png` draws the $1/\sqrt{R}$ line.

Qubit or chain level. At the *qubit* level (tutorial) each qubit is corrected by its own m_q. At the *chain* level (Teza et al., Methods, Eq. 8) all qubits of chain i are corrected by the chain's mean m_i: unbroken chains have $m_q = m_i$ for every q, and the chain-level estimate averages L_i qubits, but cannot see a bias that varies along a chain. Option `--level qubit|chain`.

Gauges. At every iteration the instance is transformed by random signs g_i: $W_{ij} \to g_i g_j W_{ij}$, $S_i \to g_i S_i$. The physics is identical, but the coupler errors now add to the qubit fields with random signs from one iteration to the next. The feedback then learns biases of the *device*, not of one instance's coupler errors. Option `--no-gauge` disables it.

6.4 What it tells about the state of the machine

The shim history (`shims/<backend>/history.json`) records per iteration: rms and max $|m_q|$, rms chain magnetisation, rms and max offset, chain-break fraction, best energy, QPU time, chip-graph fingerprint, problem id. Read it as follows:

- rms m_q at the first iteration well above $1/\sqrt{R}$: the device is biased for this embedding; the excess is its size;
- rms m_q falling to $1/\sqrt{R}$: the bias is corrected; the offsets Φ_q are a map of the device's flux offsets on the qubits used;
- rms m_q not falling: α_Φ too small (raise it) or bias drifting faster than the feedback;
- offsets of a later session very different from today's, or a different `graph_sha1`: the chip was recalibrated; shim again;
- `--alpha-phi 0`: a pure diagnostic, nothing changed, only the biases measured.

The offsets are stored in `shims/<backend>/fbos.json` and, with the default `--apply`, used by every later `qpu` call of that back-end (`flux_biases`, `flux_drift_compensation=True`); each result records how many offsets were applied and their rms (`info.flux_biases`).

6.5 Commands and cost

```
python main_simulation.py shim --out results/full2 --backend qpu:zephyr:clique  --N 64 --chec
python main_simulation.py shim --out results/full2 --backend qpu:zephyr:clique  --N 64 --iter
python main_simulation.py shim --out results/full2 --backend qpu:pegasus:clique --N 64 --iter
```

One iteration with 100 reads at $N = 64$ is ≈ 15.8 ms $+ 100 \times 0.2$ ms ≈ 37 ms; 30 iterations ≈ 1.1 s per QPU. The offsets are learned for the qubits of the clique embedding of that N; larger N use a superset of qubits, and the qubits not shimmed get no offset. To shim at the largest size, run at $N = 104$ (Zephyr) or $N = 160$ (Pegasus), $\approx$1.5–2 s each.

7 Recommended order with the remaining budget

1. `coupling-scan` with $p \in \{0.75, 1, 1.414, 2, 3, 4, 6\}$ at $N = 8, 16, \ldots, 48$ and 104, both QPUs: $7 \times 7 \times 2 = 98$ calls of $\approx 15.8\ \text{ms} + 200 \times 0.2\ \text{ms} \approx 55$ ms, ≈ 5.4 s. Expect the chosen p to rise from ≈ 2 at small N to ≈ 3–4.
2. Shimming at the largest N of each QPU, 30 iterations: ≈ 3–4 s in total.
3. Optionally the same scan at $N = 104$ with `--extended-j-range` (a second campaign directory), $p \in \{3, 4, 6\}$: < 1 s.
4. The main run with the chosen table, shims applied.